

RayGlides 3W/4W EV Kit Cooling Controller - Complete Schematic (Optimized for ESP32-S3-N16R8)

This document describes the optimized schematic design, net names, and connectivity details for the RayGlides Cooling Controller PCB.

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1. ASCII Circuit Schematics

A. Power Input and Back-Power Safety Stage

[illegible]

* **Note***: D6 prevents back-feeding 5V from the vehicle's 48V-to-5V buck regulator into the host PC's USB port when programming the module on the vehicle.

B. ESP32-S3-N16R8 Native USB-C Interface

VBUS									
?									
[J5: USB-C Connector]									
A6/B6 (D+)		A7/B7 (D-)		A1/B12 (GND)		A5 (CC1)		B5 (CC2)	
?		?		?		?		?	
????????		????????		?		[R8:5.1k]		[R9:5.1k]	
?	?	?	?	?	?	?	?	?	?
[ESD: USBLC6-2SC6]				?	?	????????????			
?	?	?	?	?	?	?			
[R10: 22R]		[R11: 22R]		?	?	?			
?	?	?	?	?	?	?			
ESP32_GPIO20	????	?	?	?	?	?			
ESP32_GPIO19	????????????	?	?	?	?	?			
GND	PWR	??	GND	PWR					

* *Note*: CC1/CC2 pull-down resistors (5.1k) configure the Type-C interface to request 5V from host power supplies. USBLC6-2SC6 clamps high-speed ESD transients.

C. Integrated CAN Bus (TJA1050) Interface

+5V

?				
[TJA1050]				
TXD (Pin 1)	RXD (Pin 4)	CANH (Pin 7)	CANL (Pin 6)	GND (Pin 2)
?	?	?	?	?
ESP32_GPI09	????	?	?	?
ESP32_GPI010	????????????????	??????[R12:60R]	??????	?
		?	?	?
		?	[C12:4.7nF]	?
		?	?	?
		??????[R13:60R]	????????????	
		?	?	
		J4.1 (CANH)	J4.2 (CANL)	
			?	
GND_PWR	??			GND_PWR

* *Note*: CAN bus uses a split 120Ω termination ($60\Omega + 60\Omega$) with a 4.7nF capacitor to bypass high-frequency common-mode noise.

D. Analog Reference and RC Filter Stage (LM35)

+5V ???				
	?		?	
	[C8: 100nF]		[LM35]	
	?		VCC V_OUT (Pin 2)	
	?		?	
GND_SENS	????????????????????????????????????	????[R6: 1k]	??????	ESP32_GPI04 (ADC1_CH3)
		?	?	
		[R7: 200R]	[C9: 100nF]	
		(Pulldown)	?	
		?	[D5: TVS 3.3V]	
		?	?	
	[LM4040-1.0V]	?	?	
	V_REF Output	????????????????????????????????		GND_SENS

* *Note*: The LM4040 1.0V precision shunt reference provides a calibrated reference voltage, ensuring linear ADC conversions.

2. Component Pinout and Interconnect Netlist

Net Name	From Component (Pin)	To Component (Pin)	Description
:--- :--- :--- :---			
48V_IN	J1 Vehicle Power (1)	F1 Fuse Input (1)	Raw vehicle DC positive line
GND_VEH	J1 Vehicle Power (2)	D2 TVS Anode (2)	Ground return line from vehicle battery
Raw_EV_48V	F1 Fuse Output (2)	D1 Schottky Anode (1)	Internal fused power bus
Protected_48V	D1 Schottky Cathode (2)	L1 Inductor Input (1)	Schottky protected supply line
12V	LM5017 Buck Output (SW)	J2 Fan Output (1)	12V rail for Fan and 5V Buck input
5V	AP63205 Output (Pin 6)	LM35 VCC (Pin 1)	5V rail for Sensor and 3.3V LDO input
3.3V	AP2112K LDO Output (Pin 5)	ESP32-S3 VDD (Pin 3)	3.3V MCU logic supply rail
GND_PWR	LM5017 Ground (Pin 5)	Q1 Source (Pin 3)	Primary buck converter ground plane
GND_SENS	LM35 Ground (Pin 3)	ESP32 Analog Ground (Pin 2)	Isolated quiet analog ground return

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ESP32_GPIO18	ESP32-S3 (Pin 22 / GPIO18)	R4 Resistor Input (1)	Fan switching PWM control signal
ESP32_GPIO4	R6 Resistor Output (2)	ESP32-S3 (Pin 8 / GPIO4)	Temperature sensor ADC input line
ESP32_GPIO9	ESP32-S3 (Pin 13 / GPIO9)	TJA1050 TXD (Pin 1)	CAN TX communication line
ESP32_GPIO10	ESP32-S3 (Pin 14 / GPIO10)	TJA1050 RXD (Pin 4)	CAN RX communication line
ESP32_GPIO19	ESP32-S3 (Pin 23 / GPIO19)	R11 Resistor Output (2)	Native USB D- data line
ESP32_GPIO20	ESP32-S3 (Pin 24 / GPIO20)	R10 Resistor Output (2)	Native USB D+ data line
CANH	TJA1050 CANH (Pin 7)	J4 CAN Connector (Pin 1)	CAN high-level differential line
CANL	TJA1050 CANL (Pin 6)	J4 CAN Connector (Pin 2)	CAN low-level differential line
FAN_GATE	R4 Resistor Output (2)	Q1 Gate (Pin 1)	MOSFET gate driving node
FAN_DRAIN	J2 Fan Pin (2)	Q1 Drain (Pin 2)	MOSFET switching node
TEMP_RAW	LM35 Output (Pin 2)	R6 Resistor Input (1)	Unfiltered analog temperature signal